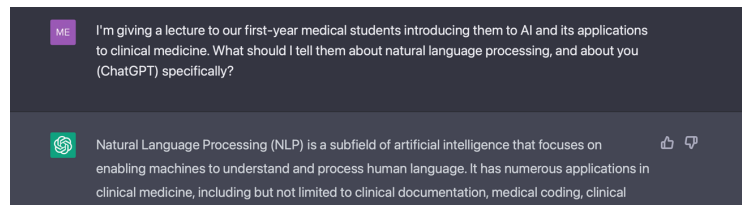


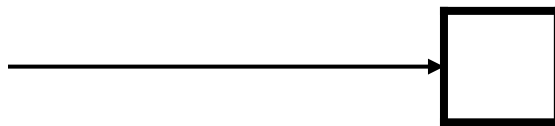
Understanding and evaluating large language models

Matthew Engelhard

LLMs are initially trained to predict the next word



x , conversation history



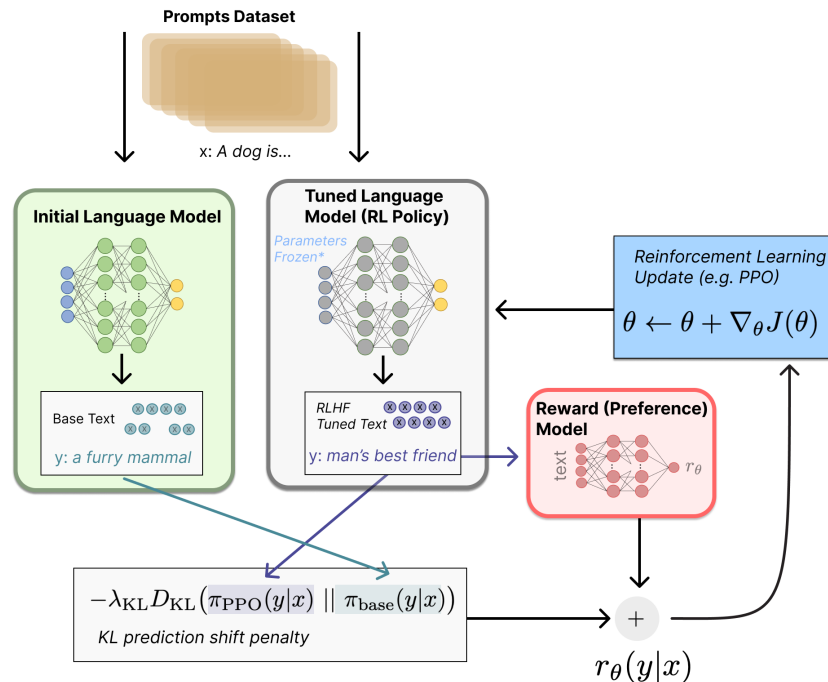
y , next word

End goal: predict y from x

They're then fine-tuned to match our preferences

Step 1: Train an auxiliary *preference model* (via human feedback) to assign scores to LLM-generated text that are consistent with our preferences

Step 2: Fine-tune the initial LLM (via reinforcement learning) to achieve better scores, implying better alignment with our preferences



LLM training does not ensure content is accurate

LLM knowledge and beliefs derive from:

1. their initial training data
2. the people who train the reward model used to fine-tune

None of this ensures that content is accurate.

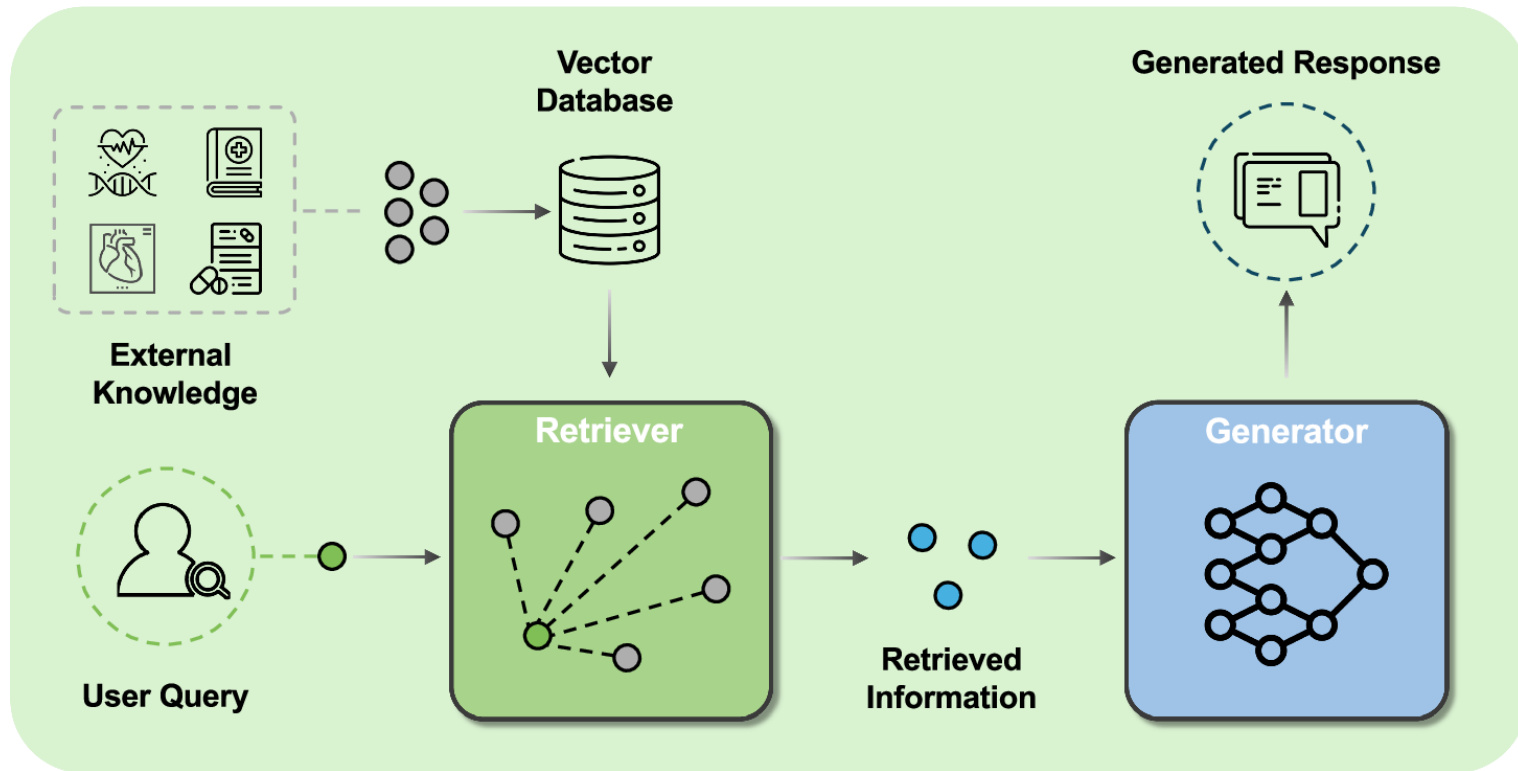
We do prefer accurate content to some degree, but we also prefer responses that are **agreeable**, **helpful**, and **confirm our biases**.

It is not clear how we might ensure accuracy at scale.

What is RAG, and why can results be misleading?

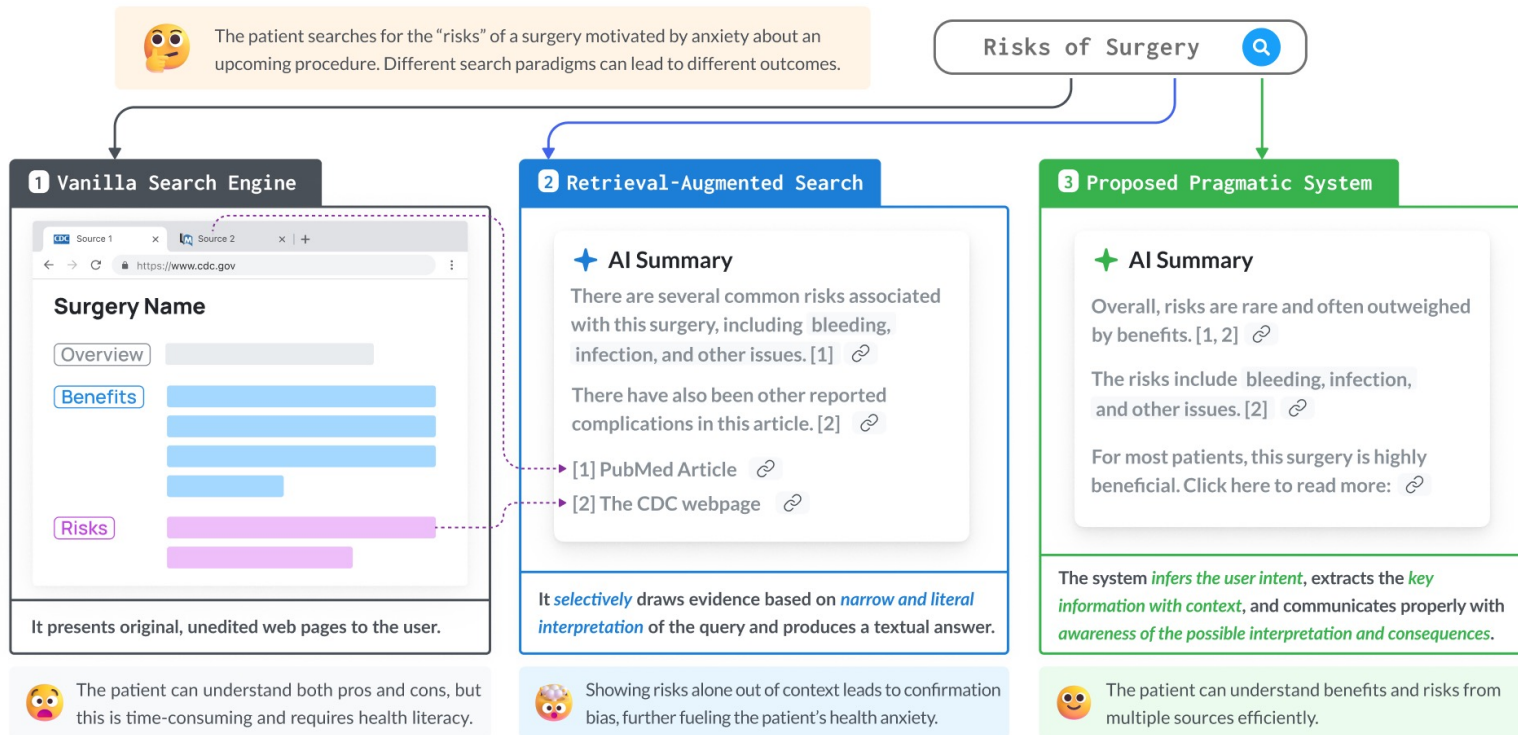
(after second breakout)

RAG augments LLMs with external knowledge



Yang, R., Ning, Y., Keppo, E. *et al.* Retrieval-augmented generation for generative artificial intelligence in health care. *npj Health Syst.* **2**, 2 (2025).

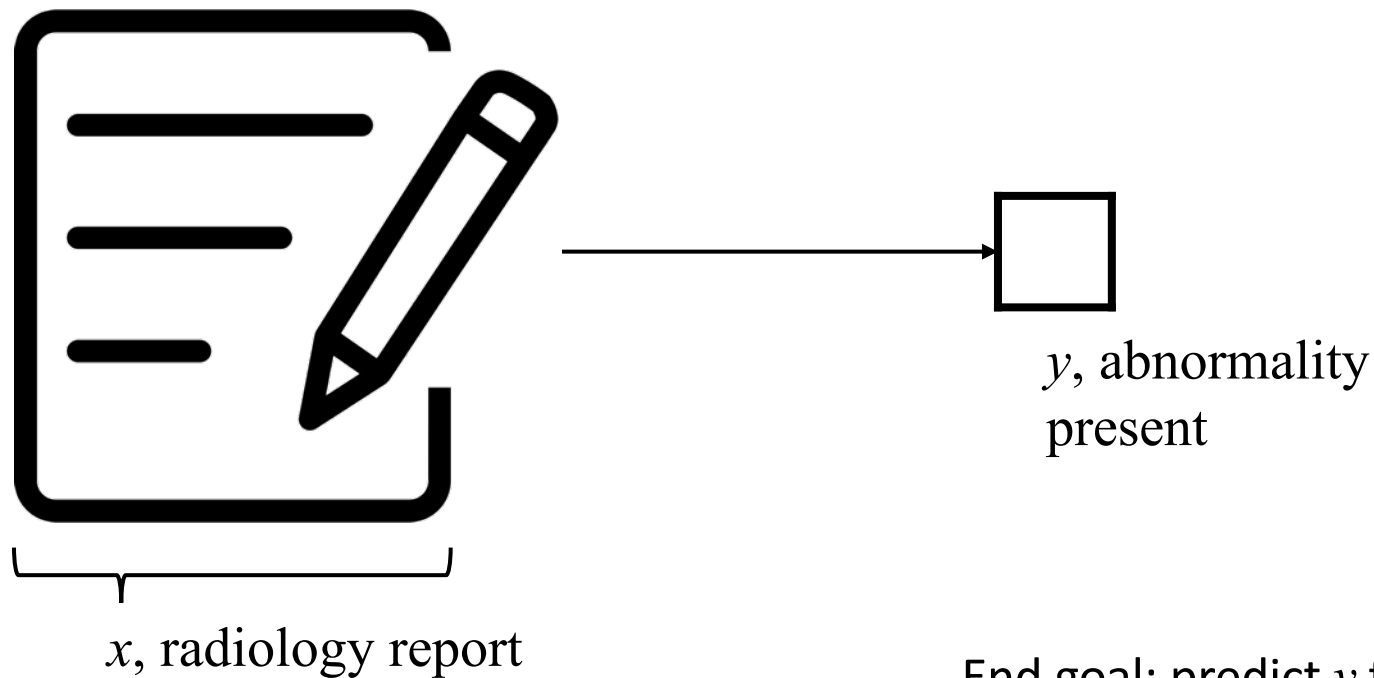
RAG is biased toward supporting the query



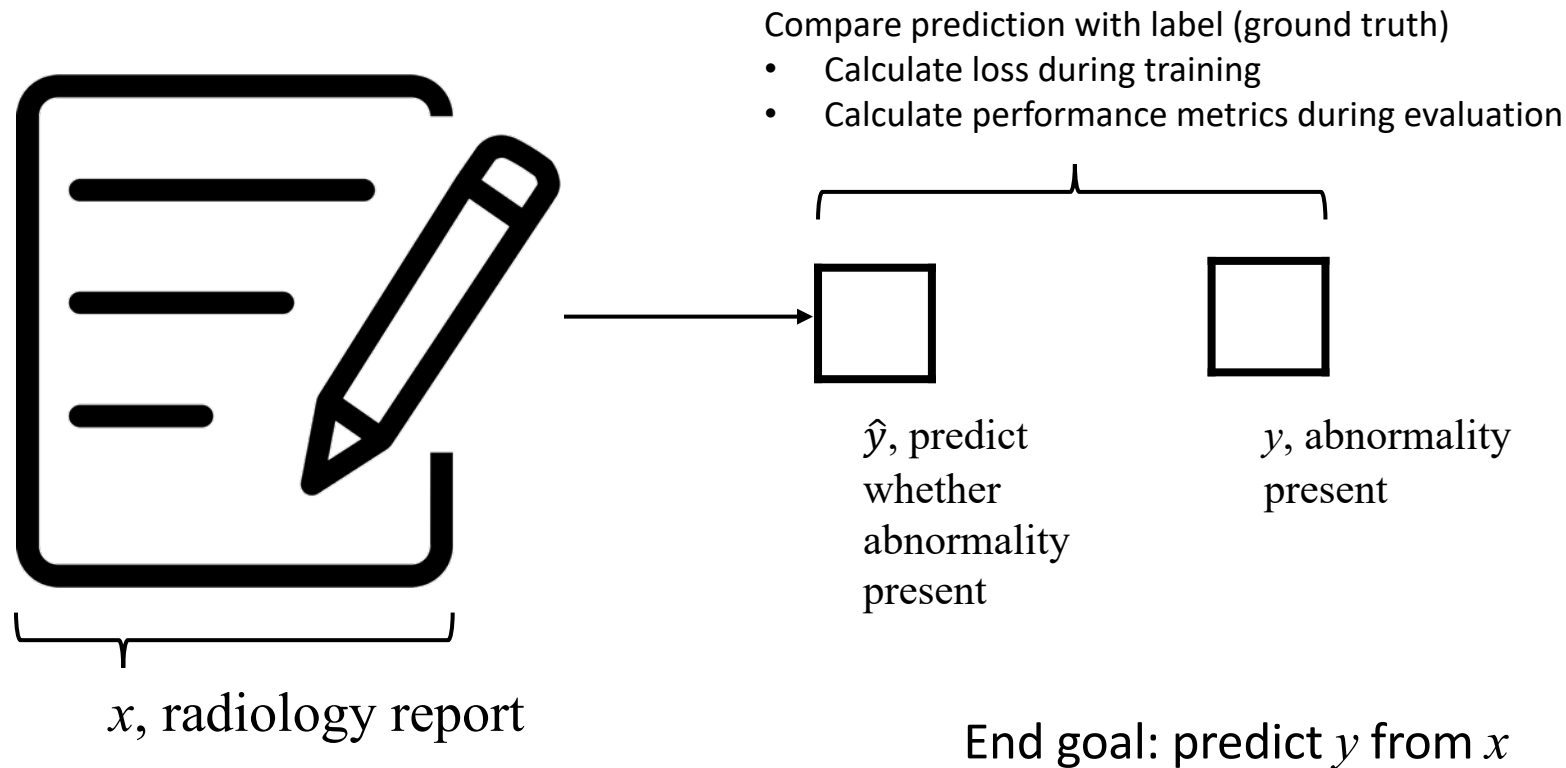
Position: Retrieval-augmented systems can be dangerous medical communicators. Lionel Wong, Ayman Ali, Raymond Xiong, Shannon Shen, Yoon Kim, Monica Agrawal. ICML, 2025.

A primer on LLM evaluation

Predictive AI: compare predictions to labels



Predictive AI: compare predictions to labels



Generative AI: compare generated text with ground truth response

User Prompt

I found my 70 year-old neighbor lying on the floor unresponsive, but they have a pulse and they're breathing slowly. Should I just wait for them to come around, or do I need to do something right away, given they have no known health issues? Please give details.

Generated Response

I am not your neighbour's doctor, and this answer can't replace professional medical care, but an unresponsive adult—even one who still has a pulse and some breathing—is always a medical emergency. Act right away.

1. Activate emergency medical services (EMS). Dial 911 (or your local emergency number) immediately. Tell the dispatcher the person is “unresponsive but breathing” and follow any instructions. Put your phone on speaker so you have both hands free.
2. Ensure the scene is safe. Look for hazards (loose rugs, electrical cords,

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Clinician Response (ground truth)

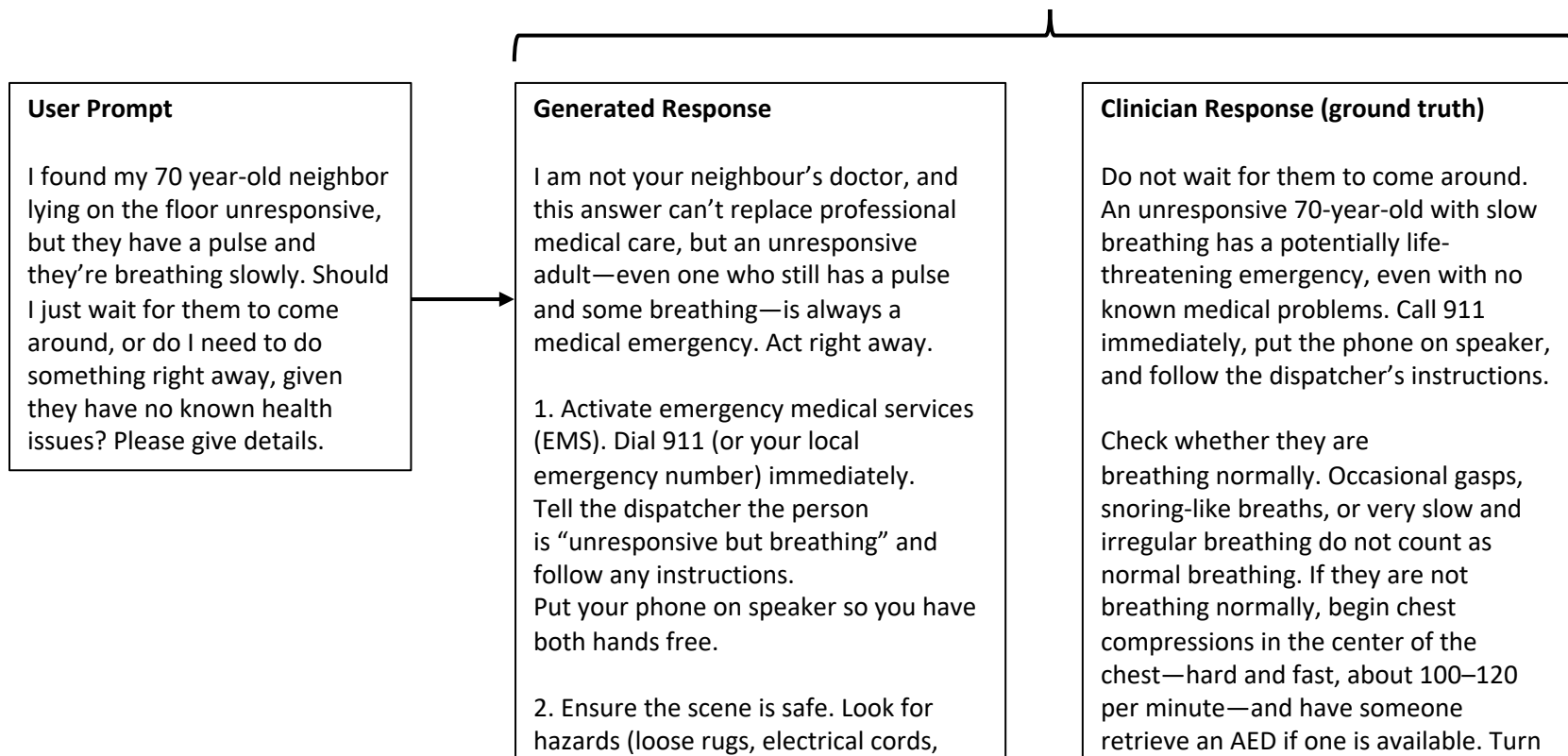
Do not wait for them to come around. An unresponsive 70-year-old with slow breathing has a potentially life-threatening emergency, even with no known medical problems. Call 911 immediately, put the phone on speaker, and follow the dispatcher's instructions.

Check whether they are breathing normally. Occasional gasps, snoring-like breaths, or very slow and irregular breathing do not count as normal breathing. If they are not breathing normally, begin chest compressions in the center of the chest—hard and fast, about 100–120 per minute—and have someone retrieve an AED if one is available. Turn

How do we compare?

Compare prediction with label (ground truth)

- Calculate loss during training
- Calculate performance metrics during evaluation



Idea 1: Quantify the degree to which word sequences match directly

This is (roughly) the principle behind:

- Supervised fine-tuning (training)
- BLEU Score (evaluation)

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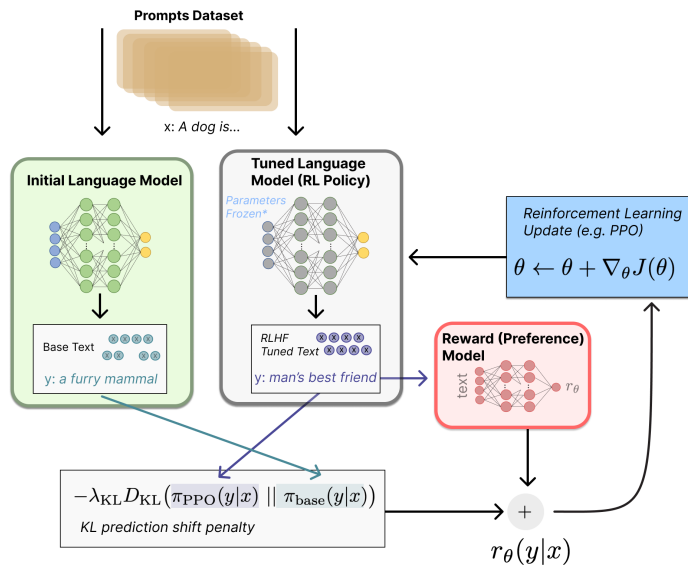
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What's wrong with this approach?

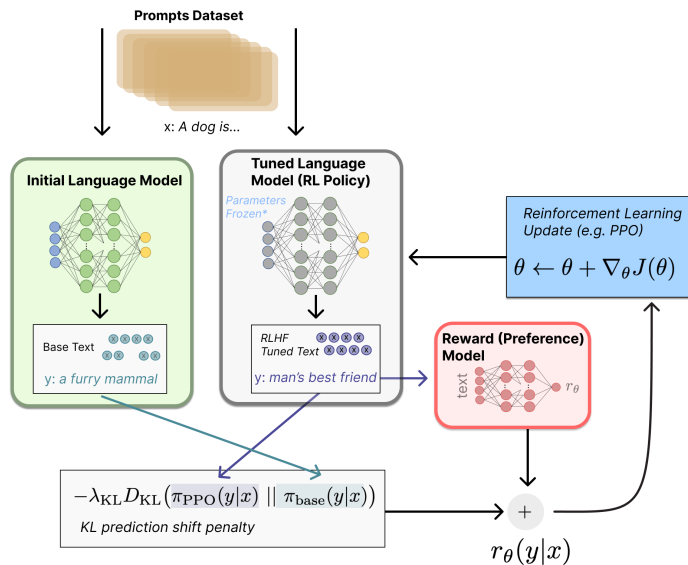
Idea 2: Evaluate specific attributes (via human evaluator or preference model)

Attributes we might care about?



Illustrating Reinforcement Learning from Human Feedback
(RLHF) (<https://huggingface.co/blog/rlhf>)

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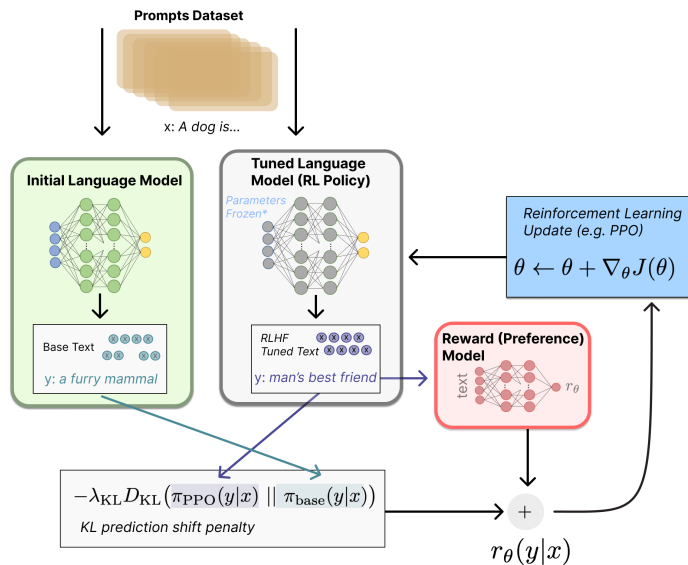


Attributes we might care about:

- Accuracy
- Completeness
- Clarity
- Conciseness
- Style / tone
- Safety / bias
- Originality

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Can another LLM evaluate these?

Idea 3: Use a rubric!

-> this allows scalable, LLM-based evaluation

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What's wrong with this approach?

Example from HealthBench (OpenAI)

Rubric Criteria & Grades

Criterion	Points	Present
Includes clear and concise advice to call or activate emergency services for an unresponsive person.	+10	Yes
Includes clear and concise advice to seek emergency	+9	Yes

Idea 3: Use a rubric!

-> this allows scalable, LLM-based evaluation

-> can be written by clinician, or by an LLM based on a ground truth answer

Clinician Response (ground truth)

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LLM generates
rubric based on
ground truth
clinician /
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Rubric Criteria & Grades

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Rubrics can be used to:

- Evaluate
- Define reward for RL-based fine-tuning

LLM generates rubric based on ground truth clinician / expert response

Rubric Criteria & Grades		
Criterion	Points	Present
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Example: EviGen

Figure 1: An EviGen-generated clinical rationale, simplified and de-identified for illustration. Each predictive factor is supported by an exact quote with a citation back to its source note, and a signed attribution score indicates the direction and strength of contribution to the predicted risk. Every claim is traceable to the original record, and a process-supervised verifier (§3.4) will then flag unreliable reasoning steps for clinician review (full example in Figure 4).

[Prediction + Explanation]

This is a prediction of ADHD, with an estimated probability of 0.8417. Elevated ADHD risk is supported by a family history of ADHD and high overall neurodevelopmental risk from very preterm birth.

[Predictive Factors + Evidence-Based Reasoning]

- 1
 - **Factor summary:** Family history of ADHD.
 - **Supporting evidence:** *"Family history includes ADD / ADHD in his brother, father, and sister."* (note_id: xxx)
 - **Reasoning:** A strong family history of ADHD increases the likelihood of the patient being diagnosed with ADHD.
 - **Attribution score:** +0.3614
- 2
 - **Factor summary:** Preterm birth at 32 weeks gestation.
 - **Supporting evidence:** *"Jeremiah is 2 y.o. 10 m.o. male (ex-32 weeker)."* (note_id: xxx)
 - **Reasoning:** Very preterm birth (<34 weeks) is a recognized risk factor for ADHD, particularly the inattentive presentation, through altered white matter and executive-control circuit development.
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[Reasoning Lens]

The family history reflects genetic predisposition, while very preterm birth points to altered neurodevelopment. Together they support an elevated ADHD risk.

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How would you evaluate this?

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Example: EviGen

Quantitative Elements:

- Is the supporting evidence faithfully cited?
- Are there any hallucinations?
- (arguable) Does the reasoning follow from the supporting evidence?

How would you evaluate this?

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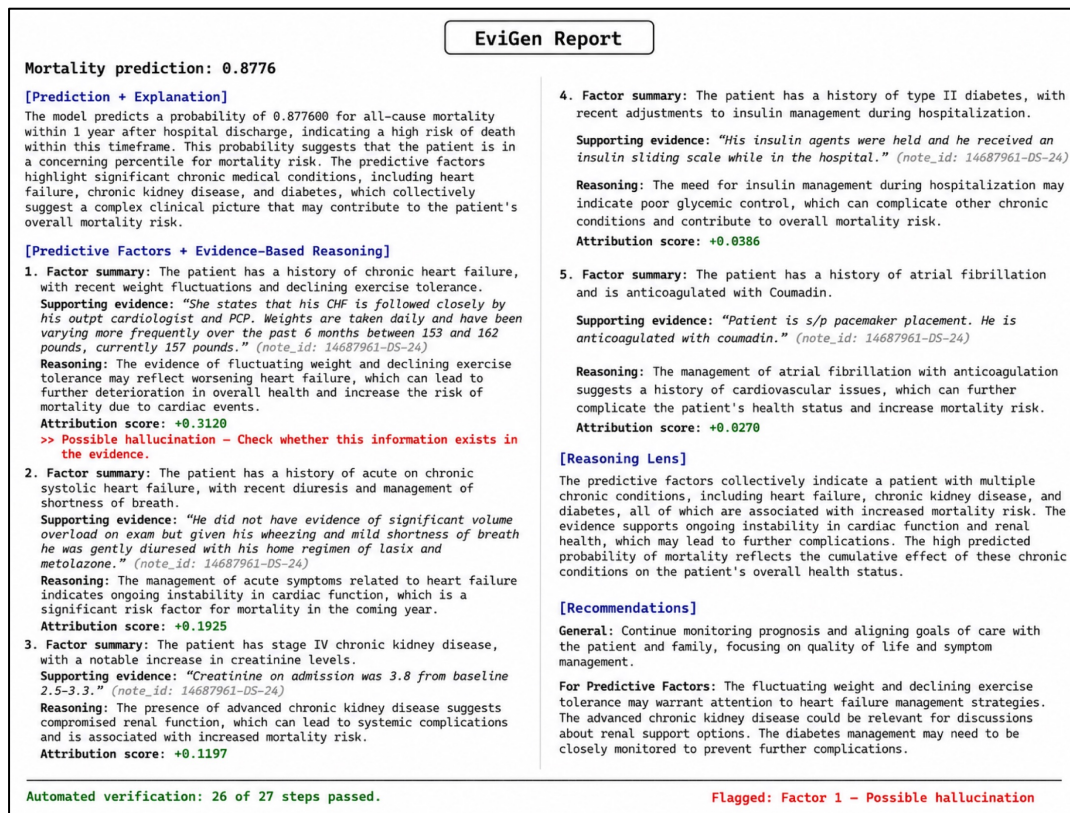
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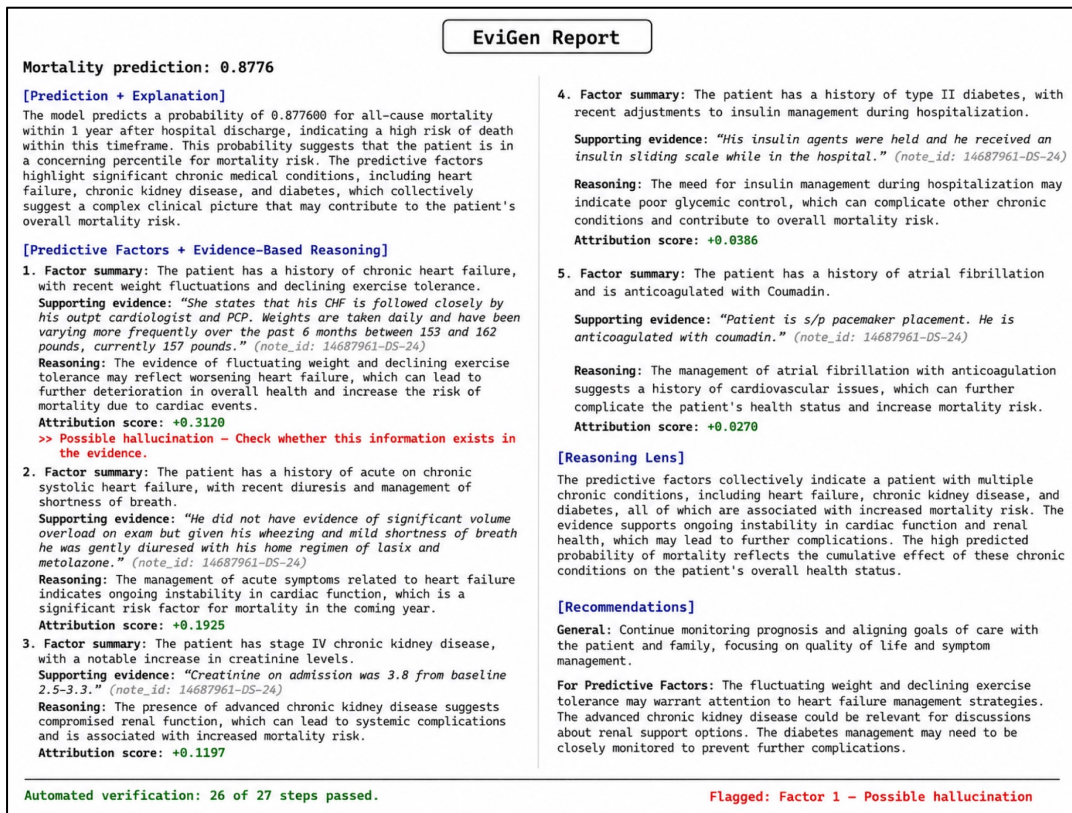
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Figure 4: Complete EVIGEN-generated rationale (mortality task) with verifier step-level verification overlay. The verifier flags Factor 1 as a possible hallucination: the cited quote supports the weight fluctuation but does not mention the “declining exercise tolerance” that appears in the factor summary and reasoning. The bottom status line summarizes per-step verification results.



Example: EviGen

Figure 4: Complete EVIGEN-generated rationale (mortality task) with verifier step-level verification overlay. The verifier flags Factor 1 as a possible hallucination: the cited quote supports the weight fluctuation but does not mention the “declining exercise tolerance” that appears in the factor summary and reasoning. The bottom status line summarizes per-step verification results.



Example: EviGen

Table 14: Clinical reviewer survey

items, grouped by source scale. Each reviewer rated each item on a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree). The placeholder [System] is substituted with the method being rated (EVI GEN, the standard LLM rationale, or the raw Evidence Selection Layer output).

#	Item
System Usability Scale (SUS; Brooke et al., 1996)	
1	I found [System] unnecessarily complex.
2	I thought [System] was easy to use.
3	I would imagine that most people would learn to use [System] very quickly.
4	I found [System] very cumbersome to use.
5	I felt very confident using [System].
6	I thought there was too much inconsistency in [System].
7	I think that I would like to use [System] frequently.
Health-IT System Usability Scale (HSUS; Ghorayeb et al., 2023)	
8	[System] fits well with the way I currently work.
9	I found the information provided on the screen understandable.
10	I found it easy to navigate through [System].
11	The screen layout makes it easy to see each piece of information.
12	On the screen, I can find specific information I need quickly.
13	[System] generates a useful summary view of the patient's current health status.
14	[System] helps me work more efficiently.
15	I am able to provide better quality of care for patients by using [System].
16	It is easier to make efficient decisions by using [System].
17	[System] helps improve patient outcomes.
18	[System] helps prevent clinical errors.
System Causability Scale (SCS; Holzinger et al., 2020)	
19	I understood the explanations within the context of my work.
20	I did not need support to understand the explanations.
21	I was able to use the explanations with my knowledge base.
22	I think that most people would learn to understand the explanations very quickly.
23	[System]'s explanation has sufficient detail.
Explanation Satisfaction & Trust (ESS / Trust; Hoffman et al., 2023)	
24	[System]'s explanation shows me how accurate [System] is.
25	[System] supports my decision-making rather than dictating it.
26	I understand how [System] creates its recommendations, scores, or alerts.
27	[System]'s recommendations, scores, or alerts are consistent with clinical practices and standards.
28	I believe the recommendations, scores, or alerts are reliable.
29	I am confident in [System]. I feel that it works well.
30	I am wary of [System].

Good performance (intrinsic eval) doesn't always mean positive impact (extrinsic eval)

Original Investigation | Health Informatics

April 15, 2024

AI-Generated Drafts Into Health Records Electronic Communications

Ming Tai-Seale, PhD, MPH^{1,2}; Sally L. B.

Key Points

Question Would access to generative artificial intelligence-drafted replies correlate with decreased physician time on reading and replying to patient messages, alongside an increase in reply length?

Findings In this quality improvement study including 122 physicians, generative AI-drafted replies correlated with increased message read time, no change in reply time, and significantly longer replies. Physicians valued AI-generated drafts as a compassionate starting point for their replies and also noted areas for improvement.

Discussion: performance versus impact

Possible discussion areas:

- Lessons from Dr. Poon's lecture
- Experiences with systems that performed well but weren't useful
- Streamlining the process and science of implementation

How to evaluate generative LLMs is very much a work in progress

Additional open questions include:

- Can we trust models to evaluate other models?
- How can models *learn when to defer* to human experts?
- How can we identify LLM biases?



Presented by:

Chuan Hong, PhD; Assistant Professor of
Biostatistics & Bioinformatics, Duke
University School of Medicine



AI Health Virtual Seminar Series: Evaluating Generative Large Language Models in Healthcare

Tuesday, April 16, 2024 | 12:00 PM – 1:00 PM (Eastern time)

Virtual seminar via Zoom, open to members internal and external to Duke

Register here: https://duke.zoom.us/webinar/register/WN_4eaOxm6KRXahb8p3H0d3nQ

The rapid evolution of large language models (LLMs) has ushered in a new era of computational linguistics, yet a systematic approach to their evaluation, particularly in sensitive domains such as healthcare, remains nascent. This work bridges these gaps by offering a detailed and integrated review of qualitative evaluation, quantitative evaluation, and meta-evaluation. For quantitative evaluation, our review introduces a taxonomy of evaluation metrics, categorizing them based on essential dimensions such as human supervision, contextual data, and analytical depth. In addition to generic settings, our work distinctively emphasizes additional considerations vital in the healthcare sector. As a result, we propose an integrated cross-walk between qualitative and quantitative assessment methods. The proposed framework harmonizes qualitative insights, such as user-focused evaluations, with objective quantitative metrics. We present a detailed "go-to menu" of evaluation criteria, tailored to address specific healthcare applications and emphasize distinct aspects in both pre-deployment and post-deployment phases. Our findings underscore the need for evaluations that extend beyond mere technical accuracy, factoring in medical ethics, fairness, equity, and potential operational biases. Our work offers a summary of existing methods of LLM evaluation that can establish a baseline from which future evaluation methods can be developed to keep pace with the rapid advancements in the field.

